Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	No antibodes detected. Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	An institutional review board (US) or independent ethics committee (Europe) approved study protocol and amendments at each participating center.	
Provide statement confirming informed consent obtained from study participants.	All participants provided written informed consent.	
Report on age and sex for all study participants.	**Age:**Patient Population Adult patients aged 18 to 55 years with RMS as defined by the revised (2010) McDonald diagnostic criteria for MS 20 with a relapsing course (ie, RMS or secondary progressive MS with superimposed relapses), an EDSS score between 0 and 5.5, and recent clinical or magnetic resonance imaging (MRI) activity were enrolled. eAppendix 1 in Supplement 1 presents complete inclusion and exclusion criteria. **Sex:**not detected.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	NCT02425644	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	not detected.	
Randomization	Methods Study Design OPTIMUM (NCT02425644) was a phase 3, multicenter, double-blind, active-comparator, superiority randomized clinical trial designed to compare the efficacy, safety, and tolerability of 20 mg of ponesimod vs 14 mg of teriflunomide in patients with RMS (eFigure 1 in Supplement 1 Trial Protocol in Supplement 2).	
Blinding	Secondary efficacy variables were (1) the change from baseline to week 108 in fatigue-associated symptoms, as measured by the symptom domain of the Fatigue Symptom and Impact Questionnaire-Relapsing Multiple Sclerosis (FSIQ-RMS), a validated patient-reported outcome measure 21 (eAppendix 1 in Supplement 1) (2) cumulative number of combined unique active lesions (CUALs) on MRI, defined as new Gd+ T1 lesions or new or enlarging T2 lesions (without double counting of lesions) from baseline to week 108 (assessed in a blinded fashion at a central reading center Medical Image Analysis Center, Basel, Switzerland) (3) time to 12-week confirmed disability accumulation (CDA) from baseline to the EOS, defined as an increase in the EDSS score, which was confirmed after 12 weeks (by an increase of at least 1.5 with a baseline EDSS score of 0.0, at least 1.0 with baseline EDSS score of 1.0 to 5.0, or at least 0.5 with a baseline EDSS score of 5.5 or more) and (4) time to 24-week confirmed CDA from baseline to the EOS.	
inclusion/exclusion criteria	Patient Population Adult patients aged 18 to 55 years with RMS as defined by the revised (2010) McDonald diagnostic criteria for MS 20 with a relapsing course (ie, RMS or secondary progressive MS with superimposed relapses), an EDSS score between 0 and 5.5, and recent clinical or magnetic resonance imaging (MRI) activity were enrolled. eAppendix 1 in Supplement 1 presents complete inclusion and exclusion criteria.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Not detected.	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	An institutional review board (US) or independent ethics committee (Europe) approved study protocol and amendments at each participating center.	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Not detected.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	not detected.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	Not detected.	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Not detected.	
If data are publicly available, provide accession number in repository or DOI or URL.	Not detected.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	